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Methods

A Note on Piecewise Affine Decision Rules for Robust, Stochastic, and Data-Driven Optimization

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Abstract. Multistage decision making under uncertainty, where decisions are taken under sequentially revealing uncertain problem parameters, is often essential to faithfully model managerial problems. Given the significant computational challenges involved, these problems are typically solved approximately. This short note introduces an algorithmic framework that revisits a popular approximation scheme for multistage stochastic programs and improves on it to deliver superior policies in the stochastic setting, as well as extend its applicability to robust optimization and a contemporary Wasserstein-based data-driven setting. We demonstrate how the policies of our framework can be computed efficiently, and we present numerical experiments that highlight the benefits of our method.

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1. Introduction

Over the past few decades, three paradigms have gained significant prominence in decision-making under uncertainty: (i) *robust optimization*, which assumes that the uncertain problem parameters lie in an uncertainty set, and the decision maker optimizes for the worst-case parameter realizations within this set; (ii) *stochastic programming*, which assumes that the uncertain parameters follow a known probability distribution, and the decision maker optimizes a risk measure under this distribution; and (iii) *data-driven optimization*, which assumes that historical samples of the uncertain parameters are available, and the decision maker optimizes a risk measure under the worst distribution that is deemed plausible in view of the observed samples. Of particular notice are Wasserstein-based data-driven optimization problems, where the set of plausible distributions is defined in terms of their Wasserstein distances to an empirical distribution.

In each of these paradigms, significant progress has been reported for single-stage problems, where all decisions are taken *here-and-now* before any of the uncertain parameters are observed. However, two-stage and multistage problems, in which some of the decisions can be taken *wait-and-see* after some or all of the parameter values have been revealed, continue to pose formidable theoretical and computational challenges. Indeed, even linear two-stage problems where the uncertain parameters only occur in the constraints' right-hand sides are known to be strongly NP-hard (Ben-Tal et al. 2004). This negative result has directed much existing and ongoing research toward developing tractable approximations. Constructing *decision rules*, which restrict the dependence of the wait-and-see decisions to low-dimensional function classes of the observed uncertainties, is particularly prominent in this context. Popular classes of decision rules include affine (Ben-Tal et al. 2004), segregate affine (Chen et al. 2008, Chen and Zhang 2009, Goh and Sim 2010), piecewise constant (Bertsimas et al. 2011a), piecewise affine (Vayanos et al. 2011, Georghiou et al. 2015, Ben-Tal et al. 2020, Thomä et al. 2024), and polynomial (Bampou and Kuhn 2011, Bertsimas et al. 2011b) rules, as well as their combinations (Rahal et al. 2021).

This short note builds on the previous works of Georghiou et al. (2015) and Bertsimas et al. (2023) to develop and analyze a refined class of piecewise affine decision rules for robust, stochastic, and data-driven

optimization. Our contributions can be summarized as follows.

1. We prove that the piecewise affine policies of Georghiou et al. (2015) cannot improve on simple affine policies in robust problems and that any improvements in stochastic problems are due to refinements in the objective function rather than increased flexibility in the wait-and-see decisions. This provides a theoretical explanation of an empirical observation frequently observed in the literature (Bertsimas and Georghiou 2015, Han and Nohadani 2025).

2. We develop a refined framework of piecewise affine policies that addresses this shortcoming and that is applicable to robust, stochastic, and contemporary Wasserstein-based data-driven problems. We show how these policies can be computed efficiently.

3. We prove strong theoretical approximation bounds under standard assumptions, matching or improving the best-known results in the literature.

4. We demonstrate through numerical experiments that our framework can yield significant improvements over the piecewise affine policies of Georghiou et al. (2015), as well as the data-driven affine policies of Bertsimas et al. (2023).

Although this short note focuses on refining piecewise affine decision rules within robust, stochastic, and data-driven optimization, we acknowledge several alternative paradigms that are widely studied in the broader literature on decision-making under uncertainty. Classical approaches rooted in Markov decision processes (MDPs) provide foundational frameworks for sequential decision making that are extensively documented in the operations research (OR)/management science (MS) literature (Puterman 1994). Recent methodological advances include approximate dynamic programming (Powell 2011) and reinforcement learning techniques, particularly deep reinforcement learning (François-Lavet et al. 2018), which leverage neural network approximations within the MDP framework. Additionally, alternative data-driven paradigms such as "predict-then-optimize" (Bertsimas and Kallus 2020), imitation learning (Abbeel and Ng 2004), and integrated prediction-optimization pipelines in combinatorial optimization (Baty et al. 2024) offer promising avenues for linking predictive and prescriptive models. However, these methods typically prioritize empirical performance and flexibility at the expense of interpretability, analytical tractability, and rigorous robustness or stochastic guarantees. To maintain clarity and brevity, we therefore do not further explore these alternative approaches here.

The remainder of this short note is structured as follows. We introduce our problem setting in Section 2. Section 3 develops our framework of decision rules, and Section 4 shows how the associated policies can be computed efficiently. Section 5 establishes theoretical approximation bounds under standard assumptions. We report on numerical experiments in Section 6. All proofs are relegated to the e-companion. Additionally, we provide the source codes and data sets to reproduce the numerical results in the GitHub repository accompanying this paper.

1.1. Notation

For $n \in \mathbb{N}$, we define $[n] = \{1, \ldots, n\}$ and $[n]_0 = [n] \cup \{0\}$. Boldface lowercase and uppercase characters $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ denote vectors and matrices, respectively, and x_i refers to the ith element of $\boldsymbol{x}$. We denote random variables by a tilde (e.g., $\tilde{x}$) and deterministic quantities, including realizations of random variables, without a tilde. We use $(x)_+$ to abbreviate $\max\{x, 0\}$. The vectors $\boldsymbol{e}_i$, $\boldsymbol{0}$, and $\boldsymbol{e}$ represent the unit standard basis vector, the vector of all zeros, and the vector of all ones, respectively. Depending on the context, $\boldsymbol{1}$ acts as the identity matrix or the identity function. For $a, b \in \mathbb{R}$, $a \le b$, $[a, b]$ is the closed interval from a to b, and for $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$, $\boldsymbol{a} \le \boldsymbol{b}$, $[\boldsymbol{a}, \boldsymbol{b}]$ is the hyperrectangle $\{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{a} \le \boldsymbol{x} \le \boldsymbol{b}\}$. We use $\boldsymbol{A}^+ = (\boldsymbol{A}^\top \boldsymbol{A})^{-1} \boldsymbol{A}^\top$ and F^+ as the left Moore-Penrose inverse of a matrix $\boldsymbol{A}$ and the left-inverse of an injective function F, respectively. For a family of sets $\Xi = \{\Xi_g\}_{g \in [G]}$, we use Ξ to denote both the family and the union $\Xi = \bigcup_{g \in [G]} \Xi_g$ of its sets; which interpretation is used will be clear from the context.

2. Piecewise Affine Policies via Liftings

We study multistage stochastic programs of the following form:

$$\begin{aligned} \underset{\boldsymbol{x} \in \mathcal{F}}{\text{minimize}} \quad & \mathbb{E}_{\mathbb{P}}[\boldsymbol{c}(\tilde{\boldsymbol{\xi}})^\top \boldsymbol{x}(\tilde{\boldsymbol{\xi}})] \\ \text{subject to} \quad & \boldsymbol{A}_g \boldsymbol{x}(\boldsymbol{\xi}) \le \boldsymbol{b}_g(\boldsymbol{\xi}) \quad \forall g \in [G], \boldsymbol{\xi} \in \Xi_g, \end{aligned} \tag{$\mathcal{P}(\mathbb{P}, \Xi)$}$$

where $\tilde{\boldsymbol{\xi}}$ is a random vector of problem parameters that are supported on the family of compact sets $\Xi = \{\Xi_g\}_{g \in [G]}$ and are governed by the probability distribution $\mathbb{P}$. The objective coefficients $\boldsymbol{c}(\cdot)$ and constraint right-hand sides $\boldsymbol{b}_g(\cdot)$ are affine in $\boldsymbol{\xi}$, and separate constraint sets need to be satisfied for each subset Ξ_g of the support. The problem parameters exhibit the temporal structure $\boldsymbol{\xi} = (\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T)$, where $\boldsymbol{\xi}_t$ is observed at the beginning of time stage $t \in [T]$ and where $\boldsymbol{\xi}^t = (\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_t)$ comprises the parameters observed up until (and including) stage t. Likewise, the decisions satisfy $\boldsymbol{x}(\cdot) = (\boldsymbol{x}_0, \boldsymbol{x}_1(\cdot), \ldots, \boldsymbol{x}_T(\cdot))$, where $\boldsymbol{x}_0$ comprises the here-and-now decisions and where the recourse decisions $\boldsymbol{x}_t$ must only depend on the parameter values $\boldsymbol{\xi}^t$ observed previously. This nonanticipativity requirement is captured by the constraint $\boldsymbol{x} \in \mathcal{F}$. To avoid confusion, we emphasize that $\boldsymbol{\xi}_t$ and $\boldsymbol{x}_t$ are subvectors of $\boldsymbol{\xi}$ and $\boldsymbol{x}$, whereas the scalars ξ_i and x_i refer to the ith elements of $\boldsymbol{\xi}$ and $\boldsymbol{x}$, respectively.

Problem ($\mathcal{P}(\mathbb{P},\Xi)$) unifies several commonly studied problem formulations from the literature.

Example 1 (Stochastic Programming). Multistage stochastic programs of the following form:

$$\begin{aligned} \underset{x\in\mathcal{F}}{\text{minimize}} \quad & \mathbb{E}_{\mathbb{P}}[c(\tilde{\xi})^\top x(\tilde{\xi})] \\ \text{subject to} \quad & Ax(\tilde{\xi}) \le b(\tilde{\xi}) \qquad \mathbb{P}\text{-a.s.} \end{aligned}$$

constitute a special case of Problem ($\mathcal{P}(\mathbb{P},\Xi)$) where $G=1$, Ξ_1 is the support of $\mathbb{P}$, that is, the smallest Borel-measurable closed set satisfying $\mathbb{P}(\Xi_1)=1$, and $(A_1,b_1)=(A,b)$.

Example 2 (Robust Optimization). Multistage robust optimization problems of the following form:

$$\begin{aligned} \underset{x\in\mathcal{F}}{\text{minimize}} \quad & \max_{\xi\in\Psi} c^\top x(\xi) \\ \text{subject to} \quad & Ax(\xi) \le b(\xi) \qquad \forall \xi\in\Psi \end{aligned}$$

emerge as an instance of Problem ($\mathcal{P}(\mathbb{P},\Xi)$) where $G=1$, $\Xi_1=\Psi$, and $\mathbb{P}$ is any distribution supported on Ψ, if we (i) augment the here-and-now decisions x_0 by an epigraphical variable $\tau\in\mathbb{R}$ that is minimized in the objective, and (ii) identify (A_1,b_1) with (A,b) augmented by the epigraphical constraint $c^\top x(\xi)-\tau\le 0$.

Example 3 (Data-Driven Optimization). The multistage data-driven optimization problem

$$\begin{aligned} \underset{x\in\mathcal{F}}{\text{minimize}} \quad & \sup_{\mathbb{P}\in\mathcal{B}_\epsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{P}}[c^\top x(\tilde{\xi})] \\ \text{subject to} \quad & Ax(\tilde{\xi}) \le b(\tilde{\xi}) \quad \mathbb{P}\text{-a.s.},\ \forall \mathbb{P}\in\mathcal{B}_\epsilon(\hat{\mathbb{P}}) \end{aligned}$$

considers the type-∞ Wasserstein ball $\mathcal{B}_\epsilon(\hat{\mathbb{P}})=\{\mathbb{P}\in\mathfrak{P}(\mathbb{R}^N) : \mathrm{d}_\infty^{\mathrm{W}}(\mathbb{P},\hat{\mathbb{P}})\le\epsilon\}$ with Borel probability measures $\mathfrak{P}(\mathbb{R}^N)$ on $\mathbb{R}^N$, Wasserstein distance

$$\mathrm{d}_\infty^{\mathrm{W}}(\mathbb{P},\mathbb{Q}) = \left\{ \Pi-\text{ess sup}\|\tilde{\xi}-\tilde{\psi}\| : \begin{bmatrix} \Pi \text{ is a joint distribution over } \tilde{\xi} \\ \text{and } \tilde{\psi} \text{ with marginals } \mathbb{P} \text{ and } \mathbb{Q} \end{bmatrix} \right\},$$

and ground metric $\|\cdot\|$, where the empirical distribution $\hat{\mathbb{P}}=\frac{1}{G}\sum_{g\in[G]}\hat{\xi}_g$ is supported on the historical samples $\hat{\xi}_1,\ldots,\hat{\xi}_G$ (Mohajerin Esfahani and Kuhn 2018, Blanchet et al. 2019, Gao and Kleywegt 2023). Bertsimas et al. (2023) show that this problem admits the equivalent reformulation

$$\begin{aligned} \underset{x\in\mathcal{F}}{\text{minimize}} \quad & \frac{1}{G}\sum_{g\in[G]} \max_{\xi\in\Xi_g} c^\top x(\xi) \\ \text{subject to} \quad & Ax(\xi)\le b(\xi) \qquad \forall g\in[G], \xi\in\Xi_g, \end{aligned}$$

where $\Xi_g=\{\xi : \|\xi-\hat{\xi}_g\|\le\epsilon\}$ are perturbation sets around the historical samples. The problem therefore emerges as a special case of Problem ($\mathcal{P}(\mathbb{P},\Xi)$), where $\mathbb{P}$ is any distribution supported on Ξ, if we (i) augment the here-and-now decisions x_0 by the epigraphical variables $\tau_g\in\mathbb{R}$ whose average is minimized in the objective, and (ii) identify each (A_g,b_g), $g\in[G]$, with (A,b) augmented by the epigraphical constraint $c^\top x(\xi)-\tau_g\le 0$.

Problem ($\mathcal{P}(\mathbb{P},\Xi)$) is NP-hard already when its objective function is deterministic, $T=2$, $G=1$, and Ξ_1 is polyhedral (Guslitser 2002). A widely applied conservative approximation replaces the wait-and-see decisions of $x(\xi)$ with affine decision rules; that is, it imposes an affine dependence of $x_t(\xi^t)$ on ξ^t for all $t\in[T]$ (Ben-Tal et al. 2004, Kuhn et al. 2011). Over the last two decades, this approximation has been refined to different classes of segregated affine (Chen et al. 2008, Chen and Zhang 2009, Goh and Sim 2010) and piecewise affine (Vayanos et al. 2011, Georghiou et al. 2015) decision rules. Here, the key idea is to lift the uncertain parameters $\xi\in\mathbb{R}^N$ to a higher-dimensional parameter vector $\xi'\in\mathbb{R}_+^{N'}$ and subsequently solve an affine decision rule approximation in the lifted space, which corresponds to a nonlinear decision rule approximation in the original space. To do so, we first select embedding matrices E_t that map the original uncertainties ξ^t up to stage t into stage t uncertainties $\theta_t=E_t\xi^t$, $t\in[T]$. We require each embedding $E^t=(E_1\cdots E_t)$ up to stage t to be information preserving, that is, to be an injective map of ξ^t to $\theta^t=(\theta_1,\ldots,\theta_t)$. Typical embeddings include rotations, scalings, averages, and differences. The uncertain parameters $\theta=\theta^T$ then reside in the embedded support $\Theta=E\Xi=\{E\xi : \xi\in\Xi\}\subseteq\mathbb{R}^I$, where $E=E^T$. Next, we select for each scalar component θ_i of θ the J_i-1 breakpoints

$$\begin{aligned} \min_{\theta\in\Theta}\theta_i = \underline{\theta}_i = z_{i0} < z_{i1} < z_{i2} < \ldots \\ < z_{i,J_i-1} < z_{iJ_i} = \overline{\theta}_i = \max_{\theta\in\Theta}\theta_i. \end{aligned}$$

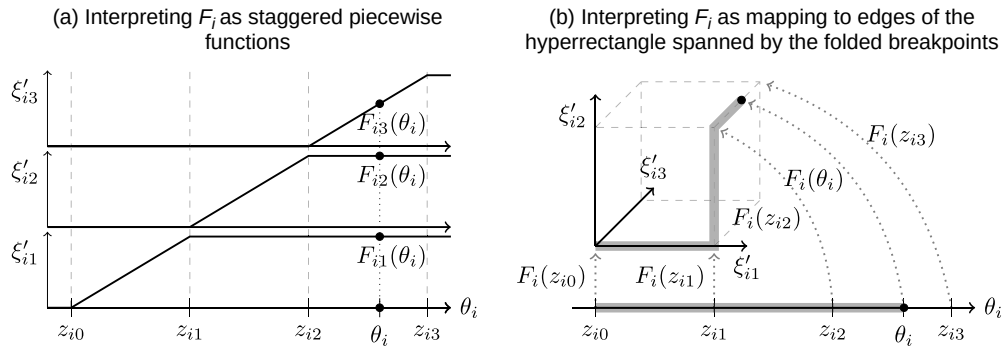
Here, z_{i0} and z_{iJ_i} are the lower and upper bounds of the embedded support Θ and not breakpoints. We then define the nonlinear folding operator F component-wise via

$$F_{ij}(\theta_i) = \begin{cases} 0 & \text{if } \theta_i < z_{i,j-1}, \\ \theta_i - z_{i,j-1} & \text{if } \theta_i\in[z_{i,j-1},z_{ij}], \\ z_{ij}-z_{i,j-1} & \text{if } \theta_i > z_{ij} \end{cases}$$

for every $i\in[I]$ and $j\in[J_i]$.

The folding operator F piecewise linearly folds each component θ_i to J_i lifted components in a staggered manner. We denote by $F_i(\theta_i)=(F_{i1}(\theta_i),\ldots,F_{iJ_i}(\theta_i))\in\mathbb{R}^{J_i}$ the subvector of $F(\theta)$ corresponding to the folding of component θ_i. Geometrically, F_i maps θ_i to an edge of the J_i-dimensional hyperrectangle spanned by the folded breakpoints. Figure 1 visualizes both of these

Figure 1. Visualization of the Piecewise Affine Folding Function F_i



interpretations. By construction, we have $\underline{\theta}_i + \sum_{j'\in[j]} F_{ij'}(\theta_i) = \min\{\theta_i, z_{ij}\}$ for all $i \in [I]$ and $j \in [J_i]$. The lifted uncertainty realizations $\boldsymbol{\xi}' = F(\boldsymbol{\theta})$ reside in the lifted support $\Xi' = F(\Theta) = \{F(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$. We define the partial foldings $\boldsymbol{\xi}'_t = F_t(\boldsymbol{\theta}_t)$ and $(\boldsymbol{\xi}')^t = F^t(\boldsymbol{\theta}^t)$ in the canonical way. Note that the embedding and folding operators have the left-inverses $\boldsymbol{E}^+ = (\boldsymbol{E}^\top \boldsymbol{E})^{-1}\boldsymbol{E}^\top$ and F^+ defined via $F_i^+(\boldsymbol{\xi}') = \underline{\theta}_i + \sum_{j\in[J_i]} \xi'_{ij}$, respectively. We combine the operators and their left-inverses to the lifting operator $L = F \circ \boldsymbol{E}$ and the retraction operator $R = \boldsymbol{E}^+ \circ F^+$, respectively. Finally, we define the lifted probability distribution $\mathbb{P}'$ via $\mathbb{P}'(S') = \mathbb{P}(\{\boldsymbol{\xi} \in \Xi : L(\boldsymbol{\xi}) \in S'\})$ for all Borel-measurable sets $S' \in \mathfrak{B}(\mathbb{R}^{N'})$, where $\mathfrak{B}(\mathbb{R}^{N'})$ is the Borel σ-algebra on $\mathbb{R}^{N'}$.

Using the lifted probability distribution $\mathbb{P}'$, the lifted support subsets $\Xi'_g = L(\Xi_g)$, and the retraction operator R, we define the lifted problem:

$$\begin{aligned} \underset{x'\in\mathcal{F}'}{\text{minimize}} \quad & \mathbb{E}_{\mathbb{P}'}[\boldsymbol{c}(R(\tilde{\boldsymbol{\xi}}'))^\top \boldsymbol{x}'(\tilde{\boldsymbol{\xi}}')] \\ \text{subject to} \quad & \boldsymbol{A}_g \boldsymbol{x}'(\boldsymbol{\xi}') \le \boldsymbol{b}_g(R(\boldsymbol{\xi}')) \\ & \qquad \forall g \in [G], \boldsymbol{\xi}' \in \Xi'_g, \end{aligned} \qquad (\mathcal{P}'(\mathbb{P}', \Xi'))$$

where now $\Xi' = \{\Xi'_g\}_{g\in[G]}$, and $\mathcal{F}'$ encodes the nonanticipativity requirement in the lifted space. Because the embedding and folding operators preserve information, $\mathcal{P}'(\mathbb{P}', \Xi')$ is equivalent to $\mathcal{P}(\mathbb{P}, \Xi)$ if both problems are solved in general decision rules, and $\mathcal{P}'(\mathbb{P}', \Xi')$ is an at least as tight conservative approximation as $\mathcal{P}(\mathbb{P}, \Xi)$ if both problems are solved in affine decision rules. For the remainder of the paper, we refer to the affine decision rule approximations of $\mathcal{P}(\mathbb{P}, \Xi)$ and $\mathcal{P}'(\mathbb{P}', \Xi')$ as Aff $\mathcal{P}(\mathbb{P}, \Xi)$ and Aff $\mathcal{P}'(\mathbb{P}', \Xi')$, respectively.

Unfortunately, the representation of the lifted support subsets Ξ'_g typically scales exponentially in the representation of the original sets Ξ_g, which renders Aff $\mathcal{P}(\mathbb{P}', \Xi')$ intractable in general. To this end, Georghiou et al. (2015) study the outer approximations

$$\begin{aligned} \overline{\Xi}'_g = \{\boldsymbol{\xi}' : F^+(\boldsymbol{\xi}') \in \Theta_g\} \\ \cap \operatorname{conv}(F([\underline{\boldsymbol{\theta}}_g, \overline{\boldsymbol{\theta}}_g])), \quad g \in [G], \end{aligned} \qquad (3)$$

of the lifted support subsets Ξ'_g for the special case when $G = 1$. Here, $[\underline{\boldsymbol{\theta}}_g, \overline{\boldsymbol{\theta}}_g] \subseteq [\underline{\boldsymbol{\theta}}, \overline{\boldsymbol{\theta}}]$ is the smallest hyperrectangle containing $\Theta_g = \boldsymbol{E}\Xi_g$. Intuitively, the first set in (3) relaxes the sufficient condition $\boldsymbol{\xi}' \in \Xi'_g = F(\Theta_g)$ to the necessary condition $F^+(\boldsymbol{\xi}') \in F^+ \circ F(\Theta_g) = \Theta_g$. Because F^+ is affine, this set is efficiently representable whenever Θ_g is. The second set in (3) is the convex hull of the folded bounding box of the embedded support, which has a polyhedral representation as per the following generalization of Georghiou et al. (2015, Lemma 4.3).

Lemma 1. *The convex hull of the folded bounding box of the embedded support satisfies*

$$\begin{aligned} &\operatorname{conv}(F([\underline{\boldsymbol{\theta}}_g, \overline{\boldsymbol{\theta}}_g])) \\ &= \Bigg\{ \boldsymbol{\xi}' : F_{ij}(\underline{\theta}_{gi}) \le \xi'_{ij} \le F_{ij}(\overline{\theta}_{gi}) \quad \forall i \in [I], j \in [J_i], \\ &\qquad \frac{\xi'_{ij} - F_{ij}(\underline{\theta}_{gi})}{F_{ij}(\overline{\theta}_{gi}) - F_{ij}(\underline{\theta}_{gi})} \ge \frac{\xi'_{i,j+1} - F_{i,j+1}(\underline{\theta}_{gi})}{F_{i,j+1}(\overline{\theta}_{gi}) - F_{i,j+1}(\underline{\theta}_{gi})} \\ &\qquad\qquad \forall i \in [I], j \in [J_i] \\ &\qquad\qquad \text{with } \underline{\theta}_{gi} < z_{ij} < \overline{\theta}_{gi} \Bigg\}. \end{aligned}$$

Lemma 4.3 of Georghiou et al. (2015) relies on the assumption that the bounding box $[\underline{\boldsymbol{\theta}}_1, \overline{\boldsymbol{\theta}}_1] = [\underline{\boldsymbol{\theta}}, \overline{\boldsymbol{\theta}}]$ aligns with the breakpoints, which implies that conv $(F([\underline{\boldsymbol{\theta}}, \overline{\boldsymbol{\theta}}]))$ coincides with the convex closure of the vertices of the hyperrectangles visualized in Figure 1(b). This does not hold for $G > 1$, which is covered by Lemma 1, however, because in that case, the bounding boxes $[\underline{\boldsymbol{\theta}}_g, \overline{\boldsymbol{\theta}}_g]$ may not be aligned with the breakpoints. As a consequence, the bounds $\underline{\boldsymbol{\theta}}_g, \overline{\boldsymbol{\theta}}_g$ may no longer be lifted to vertices of the hyperrectangles, but instead onto their edges. We elaborate on this in Online Appendix EC.1.

Because both sets on the right-hand side of (3) constitute outer approximations of Ξ'_g, we conclude that $\overline{\Xi}'_g$ indeed contains Ξ'_g. Replacing the lifted support subsets Ξ'_g with the outer approximations $\overline{\Xi}'_g$ thus results in a conservative approximation of problem

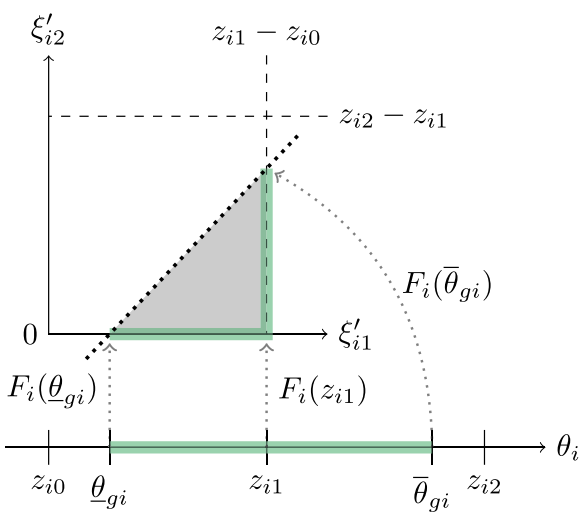
$\mathcal{P}'(\mathbb{P}', \Xi')$. Note that by linearity of F^+ and $\boldsymbol{E}$, each outer approximation $\overline{\Xi}'_g$ is convex whenever Ξ_g is. Figure 2 visualizes the convex hull of the lifted bounding box $\text{conv}(F([\underline{\boldsymbol{\theta}}_g, \overline{\boldsymbol{\theta}}_g]))$ for one dimension $i \in [I]$.

Because R is affine, every feasible solution $\boldsymbol{x}$ to Aff $\mathcal{P}(\mathbb{P}, \Xi)$ induces a feasible solution $\boldsymbol{x}' = \boldsymbol{x} \circ R$ to Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$, where $\overline{\Xi}' = \{\overline{\Xi}'_g\}_{g \in [G]}$, with the same objective value. Thus, Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ is a weakly tighter approximation of $\mathcal{P}(\mathbb{P}, \Xi)$ than Aff $\mathcal{P}(\mathbb{P}, \Xi)$. Georghiou et al. (2015) observed significant improvements of Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ over Aff $\mathcal{P}(\mathbb{P}, \Xi)$ in stochastic programs (Example 1). However, previous works did not observe any improvements of Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ over Aff $\mathcal{P}(\mathbb{P}, \Xi)$, when applying the outer approximation (3) to the robust setting (Example 2) (Bertsimas and Georghiou 2015, Han and Nohadani 2025). We next provide the theoretical justification for these observations.

Theorem 1. *For $G = 1$, there exists a subset $\overline{\Psi}' \subseteq \overline{\Xi}'$ with $R(\overline{\Psi}') = \Xi$ such that every feasible solution $\boldsymbol{x}'$ in* Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ *has a corresponding feasible solution $\boldsymbol{x}$ in* Aff $\mathcal{P}(\mathbb{P}, \Xi)$ *with $\boldsymbol{x}'(\boldsymbol{\xi}') = \boldsymbol{x}(R(\boldsymbol{\xi}'))$ for all $\boldsymbol{\xi}' \in \overline{\Psi}'$.*

Theorem 1 shows that solutions $\boldsymbol{x}'$ to Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ can be interpreted as "extensions" of solutions $\boldsymbol{x}$ to Aff $\mathcal{P}(\mathbb{P}, \Xi)$. Thus, we do not gain flexibility in terms of feasibility when solving Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ instead of Aff $\mathcal{P}(\mathbb{P}, \Xi)$. However, two solutions $\boldsymbol{x}', \boldsymbol{y}'$ that extend the same solution $\boldsymbol{x}$, that is, $\boldsymbol{x}'(\boldsymbol{\xi}') = \boldsymbol{x}(R(\boldsymbol{\xi}')) = \boldsymbol{y}'(\boldsymbol{\xi}')$ for all $\boldsymbol{\xi}' \in \overline{\Psi}'$, may differ outside $\overline{\Psi}'$ and thus yield different objective values under the lifted probability distribution $\mathbb{P}'$. In other words, any improvements observed in stochastic programming problems (Example 1) stem from the lifted probability distribution in the objective function. In robust optimization problems (Example 2), on the other hand, the lifted probability distribution does not affect the objective function, and thus Theorem 1 implies that Aff $\mathcal{P}'(\overline{\mathbb{P}}', \overline{\Xi}')$ cannot improve over Aff $\mathcal{P}(\mathbb{P}, \Xi)$ in the robust case.

Figure 2. (Color online) Visualization of $\text{conv}(F([\underline{\boldsymbol{\theta}}_g, \bar{\boldsymbol{\theta}}_g]))$



Notes. The embedded support subset $[\underline{\theta}_{gi}, \overline{\theta}_{gi}]$ (bold line segment on the abscissa) is folded into a two-dimensional nonconvex set (two perpendicular bold line segments). The convex hull of $F([\underline{\theta}_{gi}, \overline{\theta}_{gi}])$ is shown as a gray-shaded set that is bounded by the inequality $\frac{\xi'_{i1} - F_{i1}(\underline{\theta}_{gi})}{F_{i1}(\overline{\theta}_{gi}) - F_{i1}(\underline{\theta}_{gi})} \geq \frac{\xi'_{i2} - F_{i2}(\underline{\theta}_{gi})}{F_{i2}(\overline{\theta}_{gi}) - F_{i2}(\underline{\theta}_{gi})}$ (dotted line).

Corollary 1. Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ *and* Aff $\mathcal{P}(\mathbb{P}, \Xi)$ *attain the same optimal value for robust optimization problems (Example 2).*

A special case of Corollary 1 that applies to a restricted class of problems with nonnegative right-hand-side uncertainty and the absence of the embedding operator $\boldsymbol{E}$ has been proved by Thomä et al. (2024). This restricted class of problems excludes prominent applications such as the inventory management problem considered in our numerical experiments, as well as the multi-item newsvendor problem in which the absence of improvements over affine policies has most recently been observed numerically (Han and Nohadani 2025). Corollary 1 generalizes this result and implies that the piecewise affine decision rules of Georghiou et al. (2015) do not improve over affine policies in the broader class of robust optimization problems covered by Example 2.

Theorem 1 and Corollary 1 extend to the setting where the constraint matrices $\boldsymbol{A}_g$ are uncertain. As Aff $\mathcal{P}(\mathbb{P}, \Xi)$ does, in general, not admit tractable reformulations when $\boldsymbol{A}_g$ is uncertain, we defer the discussion of this more general setting to Online Appendix EC.2.

In the following, we develop tighter outer approximations $\overline{\Xi}^\star$ of the lifted support Ξ' that result in superior approximations Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}^\star)$ of $\mathcal{P}(\mathbb{P}, \Xi)$ than Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$. In particular, our approximations ensure that Aff $\mathcal{P}'(\overline{\mathbb{P}}', \overline{\Xi}^\star)$ no longer coincides with Aff $\mathcal{P}(\mathbb{P}, \Xi)$ even in the robust case.

3. Valid Cuts for the Lifted Support Ξ'

We tighten the outer approximations $\overline{\Xi}'$ of the lifted support Ξ' through cuts that are satisfied by all lifted uncertainty realizations $\boldsymbol{\xi}' \in \Xi'$ but that are violated by at least some of the additional realizations $\boldsymbol{\xi}' \in \overline{\Xi}' \setminus \Xi'$ introduced by Approximation (3). To this end, we study l_1-norm distances between embedded uncertainty realizations $\boldsymbol{\theta} \in \Theta$ and hyperrectangles $[\boldsymbol{z}^-, \boldsymbol{z}^+] \subseteq \Theta$,

$$d(\boldsymbol{\theta}, [\boldsymbol{z}^-, \boldsymbol{z}^+]) = \min_{\boldsymbol{\theta}' \in [\boldsymbol{z}^-, \boldsymbol{z}^+]} \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_1,$$

where the hyperrectangles are selected from the discrete set $\mathcal{Z} = \{[\boldsymbol{z}^-, \boldsymbol{z}^+] : \boldsymbol{z}^-, \boldsymbol{z}^+ \in Z, \boldsymbol{z}^- \leq \boldsymbol{z}^+\}$ whose vertices align to our breakpoint grid $Z = \times_{i \in [I]} \{z_{ij} : j \in [J_i]_0\}$. Note that by construction, all lifted uncertainty realizations $\boldsymbol{\xi}' \in \Xi'_g$ satisfy

$$d(F^+(\boldsymbol{\xi}'), [\boldsymbol{z}^-, \boldsymbol{z}^+]) \leq \overline{d}_g([\boldsymbol{z}^-, \boldsymbol{z}^+]),$$

with

$$\overline{d}_g([z^-, z^+]) = \max_{\boldsymbol{\theta} \in \Theta_g} d(\boldsymbol{\theta}, [z^-, z^+]).$$

Our cuts for the lifted support rely on an extension of the l_1-norm grid distance d to embedded uncertainty realizations $F^+(\boldsymbol{\xi}')$, $\boldsymbol{\xi}' \in \overline{\Xi}' \setminus \Xi'$ that we characterize next.

Observation 1. For all $\boldsymbol{\xi}' \in \overline{\Xi}'$, we have

$$d(F^+(\boldsymbol{\xi}'), [z^-, z^+]) \leq \sum_{i \in [I]} \left(\sum_{j \in J_i^+} \xi'_{ij} + z_i^- - \underline{\theta}_i - \sum_{j \in J_i^-} \xi'_{ij} \right), \quad (4)$$

where $J_i^+ = \{j \in [J_i] : z_i^+ < z_{ij}\}$ and $J_i^- = \{j \in [J_i] : z_i^- \geq z_{ij}\}$ are the index sets corresponding to the folded uncertainty components outside the box $[z^-, z^+]$. Inequality (4) is tight for $\boldsymbol{\xi}' \in \Xi'$.

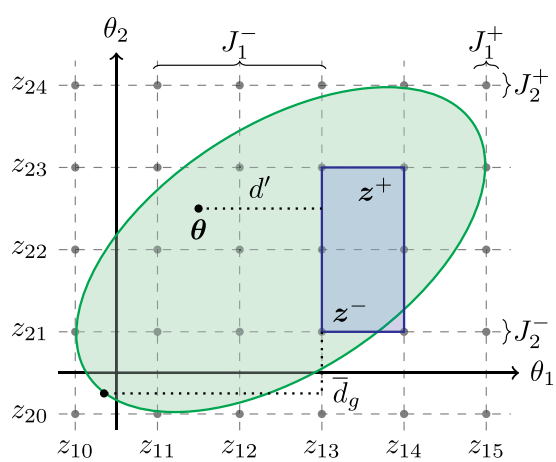
In the following, we use the notational shortcut $d'(\boldsymbol{\xi}', [z^-, z^+]) = \sum_{i \in [I]} (\sum_{j \in J_i^+} \xi'_{ij} + z_i^- - \underline{\theta}_i - \sum_{j \in J_i^-} \xi'_{ij})$ to refer to the right-hand side of (4). Observation 1 motivates us to study the refined outer approximation $\overline{\Xi}^\star = \{\overline{\Xi}_g^\star\}_{g \in [G]}$, where

$$\overline{\Xi}_g^\star = \{\boldsymbol{\xi}' \in \overline{\Xi}'_g : d'(\boldsymbol{\xi}', [z^-, z^+]) \leq \overline{d}_g([z^-, z^+]) \ \ \forall [z^-, z^+] \in \mathcal{Z}\}, \quad g \in [G], \quad (5)$$

of the lifted support subsets Ξ'_g. Figure 3 illustrates our grid distance d'.

Recall that a set $\mathcal{X} \subseteq \mathbb{R}^N$ is conic representable if $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^N : \boldsymbol{V}\boldsymbol{x} \succeq_{\mathcal{K}} \boldsymbol{d}\}$ for some $\boldsymbol{V}$, $\boldsymbol{d}$ and a proper cone $\mathcal{K}$. We next present convex reformulations of the problem

Figure 3. (Color online) Visualization of the Grid Distance



Notes. Visualization of the breakpoint grid Z (dashed), a gridpoint-induced rectangle $[z^-, z^+]$ (solid), the index sets $J_1^- = \{1, 2, 3\}$, $J_1^+ = \{5\}$, $J_2^- = \{1\}$, and $J_2^+ = \{4\}$ corresponding to the embedded uncertainty components outside the rectangle, the distance $d'(F(\boldsymbol{\theta}), [z^-, z^+])$ between an embedded uncertainty realization $\boldsymbol{\theta} \in \Theta$ and the rectangle, and the maximum distance $\overline{d}_g([z^-, z^+])$ between any point in the embedded support subset Θ_g (ellipsis) and the rectangle. To ease the exposition, we omit the arguments of d' and $\overline{d}_g$.

Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ along the lines of Georghiou et al. (2015) and our tightened approximation Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}^\star)$.

Observation 2. Assume that the support subsets Ξ_g are conic representable. Then,

(i) If Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ is feasible with a Slater point, it has the same optimal value as the convex conic program

$$\begin{aligned} \underset{\boldsymbol{\xi}', \boldsymbol{s}}{\text{maximize}} \quad & -\sum_{g \in [G], k \in [K_g]} b_{gk}(R(\boldsymbol{\xi}'_{gk})) + (s_{gk} - 1) b_{gk}(R(\boldsymbol{0})) \\ \text{subject to} \quad & \sum_{g \in [G], k \in [K_g]} \boldsymbol{A}_{gk}^\top \boldsymbol{\xi}'^\top_{gk} = -\mathbb{E}_{\mathbb{P}'}[\boldsymbol{c}(R(\tilde{\boldsymbol{\xi}}'))\tilde{\boldsymbol{\xi}}'^\top] \\ & \sum_{g \in [G], k \in [K_g]} s_{gk} \boldsymbol{A}_{gk}^\top = -\mathbb{E}_{\mathbb{P}'}[\boldsymbol{c}(R(\tilde{\boldsymbol{\xi}}'))] \\ & \boldsymbol{\xi}'_{gk} \in s_{gk} \overline{\Xi}'_g, \ s_{gk} \geq 0 \quad \forall g \in [G], k \in [K_g], \end{aligned} \quad (6)$$

where $\boldsymbol{A}_{gk}$ is the kth of the K_g rows of $\boldsymbol{A}_g$, and $b_{gk}(\boldsymbol{\xi})$ is the kth component of $\boldsymbol{b}_g(\boldsymbol{\xi})$. In particular, the constraint $\boldsymbol{\xi}'_{gk} \in s_{gk} \overline{\Xi}'_g$ is conic representable for any $g \in [G]$, $k \in [K_g]$, and $s_{gk} \geq 0$.

(ii) If Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}^\star)$ is feasible with a Slater point, it has the same optimal value as (4) tightened by the affine cuts

$$s_{gk} d'\left(\frac{1}{s_{gk}} \boldsymbol{\xi}'_{gk}, [z^-, z^+]\right) \leq s_{gk} \overline{d}_g([z^-, z^+]) \quad \forall g \in [G], k \in [K_g], [z^-, z^+] \in \mathcal{Z}. \quad (7)$$

We can readily extract optimal policies $\boldsymbol{x}'(\cdot)$ to Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}')$ and Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}^\star)$ from the shadow prices of (6), as well as its tightened counterpart presented in Observation 2(ii). We prefer to work with the dual formulations of both problems, as opposed to the primal formulations that are predominant in the literature, because the duals facilitate the delayed introduction of the cuts from (5) as discussed in the next section.

4. Efficient Cut Separation

Our refined outer approximation $\overline{\Xi}^\star$ in (5) comprises exponentially many constraints indexed by $[z^-, z^+] \in \mathcal{Z}$. Rather than including all these constraints a priori, we use a constraint generation approach to iteratively add constraints that are violated by incumbent solutions $(\boldsymbol{\xi}', \boldsymbol{s})$ to problem Aff $\mathcal{P}'(\mathbb{P}', \overline{\Xi}^\star)$. To this end, we need to solve the separation problem

$$\max_{[z^-, z^+] \in \mathcal{Z}} d'(\boldsymbol{\xi}'_{gk}/s_{gk}, [z^-, z^+]) - \overline{d}_g([z^-, z^+]) \quad (8)$$

for each $g \in [G]$ and $k \in [K_g]$ with $s_{gk} > 0$ to identify maximally violated constraints (Observation 2). Unfortunately, determining maximally violated constraints is hard in general.

Proposition 1. *The separation problem* (6) *is strongly NP-hard.*

Note, however, that $\overline{\Xi}^\star$ remains a valid outer approximation of Ξ' for any subset of the constraints in $\mathcal{Z}$, and thus we can solve the separation problem (8) heuristically

and/or terminate our constraint generation approach before all violated constraints have been included. These approaches are well established in the cutting plane literature, and we do not elaborate on them further in the following. Instead, Section 4.1 shows how the separation problem (8) can be solved exactly in polynomial time for permutation invariant uncertainty embeddings, and Section 4.2 uses these insights to approximately solve the separation problem for general embeddings.

4.1. Permutation Invariant Uncertainty Embeddings

In this section, we assume that $G = 1$, that is, we focus on stochastic and robust problems (Examples 1 and 2). Consequently, we drop the subscript g to minimize clutter. Denote by $\mathcal{S}$ the set of all signed permutation matrices in $\mathbb{R}^I$, that is, all matrices $\boldsymbol{S} \in \mathbb{R}^{I\times I}$ with elements -1, 0, and 1 that have exactly one nonzero element in each row and each column. Throughout this section, we make the following two assumptions about Z and Θ.

Assumption 1 (Permutation Invariance of Breakpoints). *The breakpoints Z are permutation invariant, that is, $\boldsymbol{S}\boldsymbol{z} \in Z$ whenever $\boldsymbol{z} \in Z$ and $\boldsymbol{S} \in \mathcal{S}$. Additionally, $\boldsymbol{0} \in Z$.*

Assumption 2 (Permutation Invariance of Embedded Support). *The embedded support Θ satisfies the following two properties:*

(i) *The embedded support Θ is permutation invariant, that is, $\boldsymbol{S}\boldsymbol{\theta} \in \Theta$ if $\boldsymbol{\theta} \in \Theta$ and $\boldsymbol{S} \in \mathcal{S}$.*

(ii) *The maximal l_1-norm of the first i components of the embedded support Θ,*

$$\eta(i) = \max_{\boldsymbol{\theta}\in\Theta} \sum_{i'\in[i]} |\theta_{i'}|,$$

has nonincreasing differences, that is, $\eta(i+1) - \eta(i) \le \eta(i) - \eta(i-1)$ for all $i \in [I-1]$.

Assumption 1 implies that the breakpoints are symmetric around the origin, as well as uniform across all axes. The assumption is relatively benign as the breakpoints are a design choice of the decision maker rather than a characteristic of the problem setting. Because of Assumption 1, the numbers of breakpoints J_i as well as the breakpoint locations z_{ij} no longer depend on the index i, which allows us to drop the index in both quantities to further reduce clutter. Assumption 2(i) applies a similar condition as Assumption 1 to the geometry of the embedded support Θ. Additionally, Assumption 2(ii) implies that as we consider additional components of $\boldsymbol{\theta}$, the incremental growth of its maximum l_1-norm over Θ is monotonically nonincreasing. Intuitively, this can be interpreted as a "diminishing returns" behavior, where each new component of $\boldsymbol{\theta}$ contributes less to increasing the maximum norm over Θ. Note that the condition on η in Assumption 2(ii) resembles the notion of submodularity. However, submodularity is typically leveraged when minimizing a function, whereas our setting involves maximization over η (Lemma EC.3 in Online Appendix EC.6), and maximizing a submodular function is known to be NP-hard in general. Instead, one may intuitively regard the nonincreasing differences condition as a discrete version of concavity, which enables efficient maximization over η. In contrast to the permutation invariance property, the nonincreasing differences of η do not appear to admit a straightforward geometric interpretation. However, by Lemma EC.1 in Online Appendix EC.6, Assumption 2(ii) can be verified numerically by solving I convex problems of the form $\max_{\boldsymbol{\theta}\in\Theta} \sum_{i'\in[i]} \theta_{i'}$ for $i \in [I]$, which is computationally tractable under the assumed convexity of Ξ. Additionally, we will see that Assumption 2 is satisfied by various prominent uncertainty sets, for example, by ellipsoids, budgets, norm balls, and their intersections. Together, Assumptions 1 and 2 allow us to dramatically reduce the search space when we determine the maximally violated constraint in (8).

Theorem 2 (Polynomial-Time Separability). *Assume that Assumptions 1 and 2 both hold.*

(i) *The separation problem (8) can be solved in time $\mathcal{O}(IJ \log I)$.*

(ii) *If, for every $j \in [J]$, there is at most one $i \in [I]$ such that $\eta(i) - \eta(i-1) \in [z_{j-1}, z_j)$, then the separation problem (8) has a square solution $[z_j\boldsymbol{e}, -z_j\boldsymbol{e}]$ for some $j \in [J/2]$.*

Although Theorem 2(i) guarantees that the separation problem (6) can be solved in polynomial time, our constraint generation approach may still require exponentially many iterations. In contrast, the condition of Theorem 2(ii) ensures that $J/2$ cuts are sufficient to solve the separation problem (6) for all possible incumbent solutions. In this case, we can directly add those $J/2$ cuts to each lifted support subset $\overline{\Xi}_g^\star$, instead of executing a constraint generation approach. As Z is a design choice, we can select it such that condition (ii) of Theorem 2 holds whenever η has strictly decreasing differences, that is, whenever $\eta(i+1) - \eta(i) < \eta(i) - \eta(i-1)$ for all i. In this context, a straightforward choice of permutation invariant breakpoints that satisfies the property of strictly decreasing differences is

$$Z = (\{\eta(i) - \eta(i-1) : i \in [I]\} \cup \{0\} \cup \{-(\eta(i) - \eta(i-1)) : i \in [I]\})^I, \tag{9}$$

or any superset thereof.

Proposition 2 (Efficient Separation for Norm-Balls). *Any l_p-norm ball embedded support $\{\boldsymbol{\theta} : \|\boldsymbol{\theta}\|_p \le \delta\}$ with $1 \le p \le \infty$ and $\delta > 0$ satisfies Assumption 2.*

Proposition 3 (Efficient Separation for Intersections). *Let $\Theta_1, \ldots, \Theta_n$ with $\eta_1, \ldots, \eta_n$ satisfy Assumption 2. Then $\Theta = \bigcap_{h\in[n]} \Theta_h$ also satisfies Assumption 2 for $\eta(i) = \min\{\eta_h(i) : h \in [n]\}$.*

Combining Propositions 2 and 3, we cover several commonly used supports, including ellipsoidal, budgeted, and intersected norm-ball supports.

4.2. General Uncertainty Embeddings

We now revisit the general case where $G \geq 1$, which includes the stochastic, robust, and data-driven settings (Examples 1–3). Whenever $G > 1$, Assumption 2 is most likely violated, and thus the polynomial-time separability of Theorem 2 no longer applies. We can, however, continue to apply the polynomial-time separation algorithms of Theorem 2 as if Assumption 2 was satisfied. We next show that this heuristic has attractive approximation guarantees.

Proposition 4. *Let Z satisfy Assumption 1, let $\Theta = \{\Theta_g\}_{g\in[G]}$ be the embedded support, and let $\Theta^{\mathrm{P}} = \{\Theta_g^{\mathrm{P}}\}_{g\in[G]}$ be such that $\Theta_g^{\mathrm{P}} \supseteq \Theta_g$ and Θ_g^{P} satisfies Assumption 2 for all $g \in [G]$. Let $\overline{\Xi}_g^{\star}$ and $\overline{\Xi}_g^{\mathrm{P}}$ be the outer approximations of Ξ_g' and $\Xi_g^{\mathrm{P}\prime}$ generated by the exponential-sized description (3).*

(i) *The outer approximation $\overline{\Xi}^{\mathrm{A}}$ of Ξ' generated by the separation algorithm of Section 4.1 satisfies*

$$\operatorname{Aff} \mathcal{P}'(\mathbb{P}', \overline{\Xi}^{\star}) \leq \operatorname{Aff} \mathcal{P}'(\mathbb{P}', \overline{\Xi}^{\mathrm{A}}) \leq \operatorname{Aff} \mathcal{P}'(\mathbb{P}', \overline{\Xi}^{\mathrm{P}}).$$

(ii) *If Θ^{P} satisfies condition* (ii) *of Theorem 2, then statement* (i) *of this proposition continues to hold if we restrict each $\overline{\Xi}_g^{\mathrm{A}}$ to square cuts $[z_j \boldsymbol{e}, -z_j \boldsymbol{e}]$ for all $j \in [J/2]$.*

Proposition 4(i) shows that applying our polynomial time separation algorithm of Section 4.1 directly to a general family of support sets Ξ provides a conservative approximation of Problem $(\mathcal{P}(\mathbb{P}, \Xi))$ that is at least as tight as first circumscribing Ξ by any outer approximation that satisfies Assumption 2 and then applying our separation algorithm to that outer circumscription. Proposition 4(ii) strengthens this result to the subclass of circumscriptions that satisfy the strictly decreasing differences condition of Theorem 2(ii).

5. Approximation Guarantees

As an immediate implication of Feige et al. (2007, Theorem 6), no finite approximation bounds exist for the general setting of $\mathcal{P}(\mathbb{P}, \Xi)$ presented in Section 2 unless $P = NP$. However, our policies admit strong approximation bounds under assumptions commonly required in the literature on theoretical bounds for adjustable decision making (Bertsimas and Goyal 2012, Ben-Tal et al. 2020, Han and Nohadani 2025).

Assumption 3. *Problem $\mathcal{P}(\mathbb{P}, \Xi)$ satisfies the following properties:*

(i) *The objective function $\boldsymbol{c}(\boldsymbol{\xi})^{\top}\boldsymbol{x}(\boldsymbol{\xi})$ is deterministic: $\boldsymbol{c}(\boldsymbol{\xi})^{\top}\boldsymbol{x}(\boldsymbol{\xi}) = \boldsymbol{c}^{\top}\boldsymbol{x}_0$.*

(ii) *The affine function $\boldsymbol{b}_g(\boldsymbol{\xi})$ in the constraint right-hand side has nonpositive coefficients for each $g \in [G]$: $\boldsymbol{b}_g(\boldsymbol{\xi}) = \boldsymbol{B}_g\boldsymbol{\xi} + \hat{\boldsymbol{b}}_g$ with $\boldsymbol{B}_g, \hat{\boldsymbol{b}}_g \leq \boldsymbol{0}$.*

(iii) *The support Ξ_g is nonnegative, full-dimensional, and down-monotone for each $g \in [G]$: $\forall \boldsymbol{\xi} \in \Xi_g, \boldsymbol{0} \leq \hat{\boldsymbol{\xi}} \leq \boldsymbol{\xi} \Rightarrow \hat{\boldsymbol{\xi}} \in \Xi_g$.*

Although significantly restricting the general setting of $\mathcal{P}(\mathbb{P}, \Xi)$, Assumption 3 still covers a wide range of relevant problem classes, including robust capacity expansion (Laguna 1998), inventory control (Bertsimas and Bidkhori 2015), and network design (Wang and Qi 2020). Given part (iii) of Assumption 3, we can always choose $\boldsymbol{E}$, such that the following also holds.

Assumption 4. *The embedded support Θ_g is convex, nonnegative, full-dimensional with $e_i \in \Theta_g$ for all $i \in [I]$, and down-monotone for each $g \in [G]$.*

Clearly, our piecewise affine policies inherit all approximation guarantees of affine policies. We further tighten these guarantees using *scaling factors* $\beta(\boldsymbol{\theta}) = \min\{\beta' \in \mathbb{R}_+ : \exists \boldsymbol{\theta}' \in \bigcap_{g\in[G]} \Theta_g, \beta'\boldsymbol{\theta}' \geq \boldsymbol{\theta}\}$ and the concept of *dominating sets*, that is, sets $\widehat{\Theta}$ such that $\max_{\boldsymbol{\theta}\in\widehat{\Theta}} \beta(\boldsymbol{\theta})$ is finite (Ben-Tal et al. 2020).

Proposition 5. *Let $\mathcal{P}(\mathbb{P}, \Xi)$ be a problem instance that satisfies Assumption 3 and let $\boldsymbol{E}$ be such that Assumption 4 holds. Then,*

$$\operatorname{Aff} \mathcal{P}'(\mathbb{P}', \overline{\Xi}^{\star}) \leq \min_{z\in Z}\{\beta(z) + \max_{g\in[G]} \overline{d}_g([\boldsymbol{0}, z])\} \cdot \mathcal{P}(\mathbb{P}, \Xi)$$

and

$$\operatorname{Aff} \mathcal{P}'(\mathbb{P}', \overline{\Xi}^{\mathrm{A}}) \leq \min_{z\in Z^{\mathrm{A}}}\{\beta(z) + \max_{g\in[G]} \overline{d}_g([\boldsymbol{0}, z])\} \cdot \mathcal{P}(\mathbb{P}, \Xi),$$

where $Z^{\mathrm{A}} \subseteq Z$ is the subset of all grid points z such that $[\boldsymbol{0}, z] \in \mathcal{Z}$ satisfies all cuts of Type (7).

Proposition 5 directly connects the approximation bounds to grid points z such that $[\boldsymbol{0}, z]$ satisfies our new cuts (7). Enforcing cuts for critical grid points secures strong guarantees, even if not all cuts were added. Moreover, choosing the right breakpoints has a significant impact on the bounds. In particular, including a single breakpoint per dimension motivated by the policies introduced by Ben-Tal et al. (2020) replicates all approximation bounds of their policies. These bounds also extend to the multistage setting and are, to the best of our knowledge, the tightest a priori bounds known thus far.

6. Numerical Experiments

We test our piecewise affine decision rules on a variant of the single-product multiperiod inventory replenishment problem studied by See and Sim (2010) and Bertsimas et al. (2023):

$$\begin{aligned}
\underset{\boldsymbol{x},\boldsymbol{y},\boldsymbol{I}}{\text{minimize}} \quad & \mathbb{E}_{\mathbb{P}}\left[\sum_{t\in[T]} c_0 y + c_t |x_t(\tilde{\boldsymbol{\xi}}^{t-1})| + h_t(I_t(\tilde{\boldsymbol{\xi}}^t))_+ + b_t(-I_t(\tilde{\boldsymbol{\xi}}^t))_+\right] \\
\text{subject to} \quad & I_t(\tilde{\boldsymbol{\xi}}^t) = I_{t-1}(\tilde{\boldsymbol{\xi}}^{t-1}) + y + x_t(\tilde{\boldsymbol{\xi}}^{t-1}) - \tilde{\xi}_t \\
& \qquad\qquad \mathbb{P}\text{-a.s.}, \ \forall t \in [T] \\
& \sum_{t\in[T]} (-I_t(\tilde{\boldsymbol{\xi}}^t))_+ \leq s \qquad \mathbb{P}\text{-a.s.} \\
& \underline{x}_t \leq x_t(\tilde{\boldsymbol{\xi}}^{t-1}) \leq \overline{x}_t \qquad \mathbb{P}\text{-a.s.}, \ \forall t \in [T] \\
& y \geq 0.
\end{aligned}$$

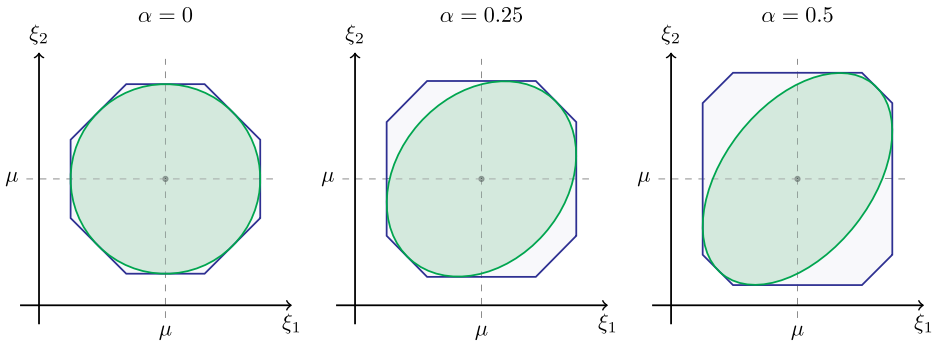
In this problem, the decision maker precommits to a single replenishment quantity y that applies to each time period $t \in [T]$ here-and-now and reactively adjusts this quantity by $x_t(\tilde{\boldsymbol{\xi}}^{t-1})$ wait-and-see on observing the uncertain customer demands $\tilde{\xi}_{t-1}$. The objective function minimizes the aggregate sum of preorder costs $c_0 y$, adjustment costs $c_t|x_t(\tilde{\boldsymbol{\xi}}^{t-1})|$, inventory holding costs $h_t(I_t(\tilde{\boldsymbol{\xi}}^t))_+$, and backlogging costs $b_t(-I_t(\tilde{\boldsymbol{\xi}}^t))_+$ across all time periods. The constraints record the inventory levels $I_t(\tilde{\boldsymbol{\xi}}^t)$, restrict the backlogged demands to at most s units, and enforce nonnegativity as well as the adjustment limits $[\underline{x}_t, \overline{x}_t]$, respectively.

In the variant studied by See and Sim (2010) and Bertsimas et al. (2023), affine decision rules yield close to optimal policies. To make the problem more challenging, we added a precommitted replenishment decision y that is dynamically adjustable by x_t and the service level constraint involving s. Other than that, we use parameters similar to See and Sim (2010) and Bertsimas et al. (2023). We set the replenishment costs to $c_0 = 0.01$, the adjustment costs to $c_t = 0.1$, the inventory holding costs to $h_t = 0.04$, the backlogging costs to $b_t = 0.2$ for $t \in [T-1]$ and $b_T = 2$, and the adjustment limits to $[\underline{x}_t, \overline{x}_t] = [-200, 200]$ for all $t \in [T]$. Further, we fix the initial inventory to $I_0(\cdot) = 0$. We assume that the demands $\tilde{\boldsymbol{\xi}}$ follow the autoregressive process $\tilde{\xi}_t = \mu + \nu(\tilde{\phi}_t + \alpha \sum_{t' \in [t-1]} \tilde{\phi}_{t'})$, $t \in [T]$, where $\mu = 200$, $\nu = \mu/\sqrt{T}$, $\alpha \in \{0, 0.25, 0.5\}$, and the constituent uncertainties $\tilde{\phi}$ are uniformly distributed over the unit two-norm ball $\{\phi : \|\phi\|_2 \le 1\}$, restricted to satisfy the limits $\tilde{\boldsymbol{\xi}}_t \in [0, 2\mu]$. This demand structure reflects independent demand shocks $\tilde{\phi}_t$ that have a permanent impact on future demands, and it is commonly used in the literature to capture long-term impacts of disruptions (Muth 1960, Graves 1999, See and Sim 2010, Bertsimas et al. 2023). We assume a service level of $s = 0.2\nu T$ and use the identity embedding $\boldsymbol{E} = \mathbf{1}$. Thus, $\Theta = \Xi$ only satisfies Assumption 2 when $\alpha = 0$, allowing us to investigate the performance of $\overline{\Xi}^{\mathrm{A}}$, when Assumption 2 is violated. Figure 4 visualizes the retraction of our new cuts and the autoregressive support Ξ for different values of α. Note that we do not display the retraction of the outer approximations $\overline{\Xi}^{\star}$, because they always retract to Ξ by construction (i.e., $R(\overline{\Xi}^{\star}) = \Xi$) and would thus make the visualization uninformative. As α increases, the figure shows that our cuts become progressively weaker.

In our experiments we benchmark our piecewise affine policies with one breakpoint at μ in each dimension (LIFT1), our piecewise affine policies with three equidistant breakpoints $\{\mu/2, \mu, 3\mu/2\}$ (LIFT3), our piecewise affine policies with three breakpoints $\{\mu + \eta(\lfloor I/2 \rfloor) - \eta(\lfloor I/2 \rfloor + 1), \mu, \mu + \eta(\lfloor I/2 \rfloor + 1) - \eta(\lfloor I/2 \rfloor)\}$ (LIFT3′), and our piecewise affine policies with all $2T-1$ breakpoints as constructed in (7) (LIFTF) against their counterparts GLIFT1, GLIFT3, GLIFT3′, and GLIFTF proposed by Georghiou et al. (2015), as well as classical affine policies (AFF). For the cut-based policies LIFT1, LIFT3, LIFT3′, and LIFTF, we only use square cuts $[z_j \boldsymbol{e}, -z_j \boldsymbol{e}]$ for all $j \in [J/2]$. These cuts are sufficient to achieve significant improvements. Online Appendix EC.11 demonstrates that dynamically adding more cuts only yields marginal benefits at the cost of significant increases in solution times.

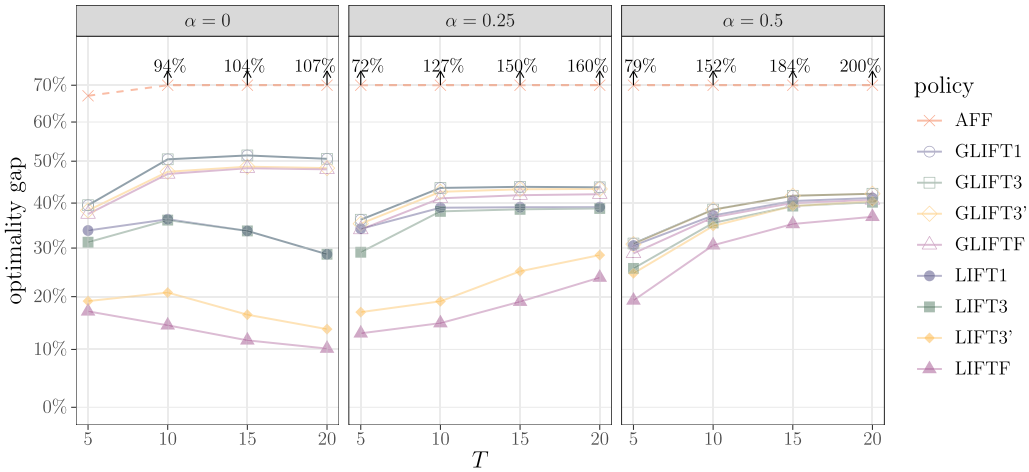
We compare all policies to dynamic programming benchmarks obtained by backward induction over a discretized grid-based state space combined with a grid search over the precommitted replenishment quantity y. The discretized state comprises the inventory level $I_t(\boldsymbol{\xi}^t)$, the accumulated backlogged demand $\sum_{t' \in [t]} (-I_{t'}(\boldsymbol{\xi}^{t'}))_+$, the summed constituent uncertainty $\sum_{t' \in [t]} \phi_{t'}$, and its squared two-norm $\|\phi^t\|_2^2$ for each $t \in [T]$. Together, the summed constituent uncertainty and its squared two-norm uniquely determine the conditional distribution of the remaining uncertainties. We discretize the inventory level and backlogged

Figure 4. (Color online) Visualization of Cuts for the Autoregressive Support



Note. Visualization of the autoregressive support Ξ (ellipsis) and the retracted cuts $R(\{\boldsymbol{\xi}' \in \mathrm{conv}(F([\underline{\boldsymbol{\theta}}, \overline{\boldsymbol{\theta}}])) : d'(\boldsymbol{\xi}', [\mu \boldsymbol{e}, \mu \boldsymbol{e}]) \le \overline{d}([\mu \boldsymbol{e}, \mu \boldsymbol{e}])\})$ (polytope) for different autoregressive factors α with a single breakpoint μ in two dimensions.

Figure 5. (Color online) Stochastic Programming



Notes. Relative objective improvements of different piecewise affine policies over affine policies for different serial demand correlations α and time horizons T. Values of AFF exceeding a 70% optimality gap are not plotted at scale to ensure the visibility of relevant effects from piecewise affine policies.

demand by $\delta = 2.5$, the summed constituent uncertainty by δ/ν, and its squared two-norm by $(\delta/\nu)^2$. This resolution corresponds to an effective demand discretization of approximately 1% of the expected value $\mathbb{E}_{\mathbb{P}}(\tilde{\xi}_t) = 200$, and numerical tests, included in the accompanying repository, indicate that further reducing δ has a negligible impact on the results. All experiments were run on an AMD EPYC 9334 2.70-GHz processor using Gurobi 11 with a single core per instance.

6.1. Stochastic Programming

Figure 5 compares the performance of the different policies against the dynamic programming solutions. Piecewise affine policies reduce the optimality gap by approximately 50% relative to their affine counterparts on symmetric instances with $\alpha = 0$, with the cuts proposed in this paper halving the remaining gap once more. For instances violating Assumption 2, the improvements of the new cuts are smaller but remain significant, still almost halving the gap for $\alpha = 0.25$ and reducing the gap by around 10% for $\alpha = 0.5$. Table 1 shows that these improvements need to be balanced against increases in computation times. Interestingly, LIFT3′ consistently outperforms LIFT3, without significant increases in computational time, indicating the importance of the breakpoint selection. In particular, breakpoints should separate the support into subdomains with significant probability mass where the optimal policy exhibits significant nonlinearities. In LIFT3′ and LIFTF, we separate the support into subdomains based on geometry. Identifying subdomains where optimal affine policies change requires domain knowledge or extensive numerical testing. In our numerical experiments, for example, we found that placing a breakpoint at the mean demand μ was beneficial for all tested piecewise affine policies. Moreover, LIFT3′ achieves similar results as LIFTF while incurring a fraction of the solution times. We thus conclude that the additional solution time from constructing breakpoints according to (9) may not be worthwhile in practice.

6.2. Robust Optimization

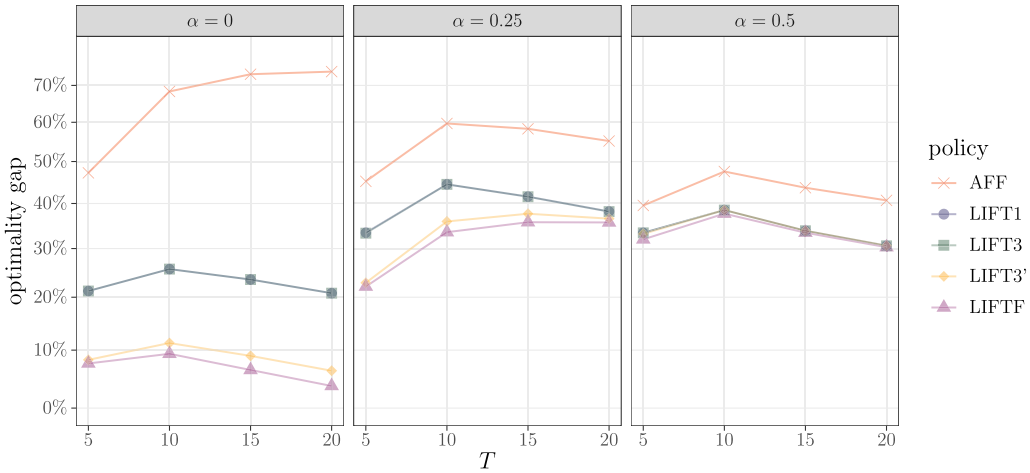
We continue to use the same autoregressive demand process, but we now consider a worst-case objective and require all constraints to be satisfied for every possible demand realization. Figure 6 reports the optimality gap improvements of our piecewise affine policies; we omit solution times, which follow a similar pattern as those for the stochastic case. We again observe significant improvements of our piecewise

Table 1. Stochastic Programming

T	AFF	GLIFT1	GLIFT3	GLIFT3′	GLIFTF	LIFT1	LIFT3	LIFT3′	LIFTF
5	0.00	0.01	0.01	0.02	0.04	0.01	0.01	0.02	0.08
10	0.02	0.04	0.09	0.10	3.34	0.04	0.09	0.08	6.30
15	0.08	0.18	0.54	0.69	34.17	0.12	0.52	0.61	81.12
20	0.24	0.47	1.20	1.44	192.49	0.31	1.18	1.59	384.49

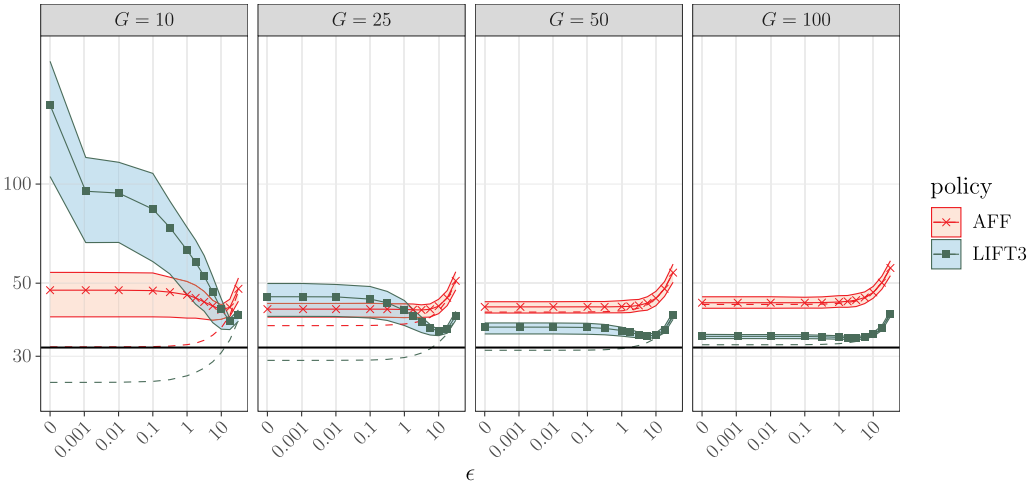
Note. Average solution times (in seconds) for the different policies from Figure 5.

Figure 6. (Color online) Robust Optimization



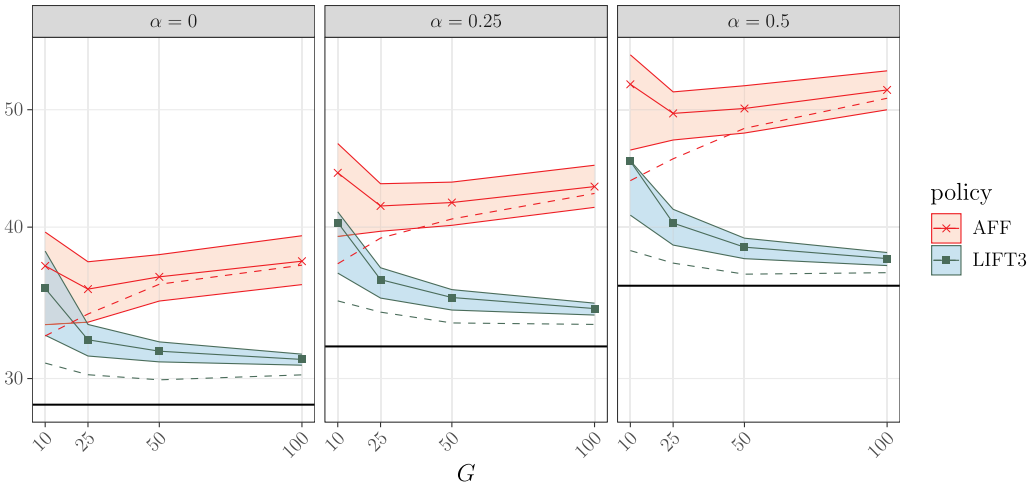
Note. Relative objective improvements of different piecewise affine policies over affine policies.

Figure 7. (Color online) Data-Driven Optimization



Notes. Out-of-sample performance of affine and our piecewise affine policies for different radii ϵ. The shaded regions represent the (20%, 80%)-confidence regions, the dashed lines report average performances on the training data, and the solid black line reports the dynamic programming solution.

Figure 8. (Color online) Data-Driven Optimization with Cross-Validation



Notes. Out-of-sample performance of affine and our piecewise affine policies where the radii ϵ are selected via fivefold cross-validation. The shaded regions, dashed lines, and solid black line have the same interpretation as in Figure 7.

Table 2. Data-Driven Optimization

	AFF		LIFT3	
G	Solution time	Violation rate	Solution time	Violation rate
10	3.84	2.8%	38.18	3.2%
25	20.82	0.7%	217.31	1.1%
50	59.81	0.3%	693.90	0.8%
100	221.57	0.0%	2,684.33	0.4%

Note. Average solution times (in seconds) and percentage of service level violations for the different policies from Figure 8.

affine policies over the affine policies, which amount to an almost 10-fold reduction of the optimality gap for larger instances with a low serial demand correlation. Again, these improvements significantly decrease for instances violating Assumption 2. Note that the piecewise affine policies of Georghiou et al. (2015) coincide with the affine policies in this experiment and are thus omitted.

6.3. Data-Driven Optimization

For the data-driven framework, we generate training sample paths $\hat{\boldsymbol{\xi}}_1, \ldots, \hat{\boldsymbol{\xi}}_G$ from the demand distribution and use the rectangular support sets $\Xi_g = \{\boldsymbol{\xi} : \|\boldsymbol{\xi} - \hat{\boldsymbol{\xi}}_g\|_\infty \leq \epsilon\}$ around each sample path $\hat{\boldsymbol{\xi}}_g$ with radii $\epsilon \in \{0\} \cup \{10^x : x = -3, -2, -1, -1/2, 0, 1/4, 1/2, 3/4, 1, 5/4, 3/2\}$. Our policy construction assumes no prior knowledge of the demand distribution. In particular, we choose the three equidistant breakpoints $\{x(\overline{\boldsymbol{\xi}} - \underline{\boldsymbol{\xi}})/4 : x = 1, 2, 3\}$ for LIFT3 within the bounding box of the empirical domain $\Xi = \bigcup_{g \in [G]} \Xi_g$. Figure 7 reports the out-of-sample performance of affine and piecewise affine policies over 10,000 uncertainty realizations and 200 randomly generated instances with demand correlation $\alpha = 0.25$ for different training set sizes G and radii ϵ, and Figure 8 reports the out-of-sample performances for different training set sizes G and serial demand correlations α if the radii ϵ in our piecewise affine policies are selected via fivefold cross-validation. Although our hyperparameter tuning is purely based on the objective values, Table 2 shows that both policies fulfill the service level constraint with high probability. Both figures demonstrate that we can expect significant performance improvements when we employ our piecewise affine policies instead of affine ones. Table 2 shows that these improvements again have to be balanced against increases in computation times, which increase by around a factor of 10. Both policies' training times increase subquadratically with the size G of the training set. The effect of α on the training times is negligible and thus omitted.

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References

Abbeel P, Ng AY (2004) Apprenticeship learning via inverse reinforcement learning. *Proc. 21st Internat. Conf. Machine Learn.* (ACM, New York), 1–8.

Bampou D, Kuhn D (2011) Scenario-free stochastic programming with polynomial decision rules. *Proc. 50th IEEE Conf. Decision Control Eur. Control Conf.* (IEEE, Piscataway, NJ), 7806–7812.

Baty L, Jungel K, Klein PS, Parmentier A, Schiffer M (2024) Combinatorial optimization-enriched machine learning to solve the dynamic vehicle routing problem with time windows. *Transportation Sci.* 58(4):708–725.

Ben-Tal A, Housni OE, Goyal V (2020) A tractable approach for designing piecewise affine policies in two-stage adjustable robust optimization. *Math. Programming* 182(1–2):57–102.

Ben-Tal A, Goryashko A, Guslitzer E, Nemirovski A (2004) Adjustable robust solutions of uncertain linear programs. *Math. Programming* 99(2):351–376.

Bertsimas D, Bidkhori H (2015) On the performance of affine policies for two-stage adaptive optimization: A geometric perspective. *Math. Programming* 153(2):577–594.

Bertsimas D, Georghiou A (2015) Design of near optimal decision rules in multistage adaptive mixed-integer optimization. *Oper. Res.* 63(3):610–627.

Bertsimas D, Goyal V (2012) On the power and limitations of affine policies in two-stage adaptive optimization. *Math. Programming* 134(2):491–531.

Bertsimas D, Kallus N (2020) From predictive to prescriptive analytics. *Management Sci.* 66(3):1025–1044.

Bertsimas D, Goyal V, Sun XA (2011a) A geometric characterization of the power of finite adaptability in multistage stochastic and adaptive optimization. *Math. Oper. Res.* 36(1):24–54.

Bertsimas D, Iancu DA, Parrilo PA (2011b) A hierarchy of near-optimal policies for multistage adaptive optimization. *IEEE Trans. Automated Control* 56(12):2809–2824.

Bertsimas D, Shtern S, Sturt B (2023) A data-driven approach to multistage stochastic linear optimization. *Management Sci.* 69(1):51–74.

Blanchet J, Kang Y, Murthy K (2019) Robust Wasserstein profile inference and applications to machine learning. *J. Appl. Probability* 56(3):830–857.

Chen X, Zhang Y (2009) Uncertain linear programs: Extended affinely adjustable robust counterparts. *Oper. Res.* 57(6):1469–1482.

Chen X, Sim M, Sun P, Zhang J (2008) A linear decision-based approximation approach to stochastic programming. *Oper. Res.* 56(2):344–357.

Feige U, Jain K, Mahdian M, Mirrokni V (2007) Robust combinatorial optimization with exponential scenarios. Fischetti M, Williamson DP, eds. *Integer Programming and Combinatorial Optimization* (Springer, Berlin), 439–453.

François-Lavet V, Henderson P, Islam R, Bellemare MG, Pineau J (2018) An introduction to deep reinforcement learning. *Foundations Trends Machine Learn.* 11(3–4):219–354.

Gao R, Kleywegt A (2023) Distributionally robust stochastic optimization with Wasserstein distance. *Math. Oper. Res.* 48(2):603–655.

Georghiou A, Wiesemann W, Kuhn D (2015) Generalized decision rule approximations for stochastic programming via liftings. *Math. Programming* 152(1–2):301–338.

Goh J, Sim M (2010) Distributionally robust optimization and its tractable approximations. *Oper. Res.* 58(4):902–917.

Graves SC (1999) A single-item inventory model for a nonstationary demand process. *Manufacturing Service Oper. Management* 1(1):50–61.

Guslitser E (2002) Uncertainty-immunized solutions in linear programming. MS thesis, Technion, Israel Institute of Technology, Haifa.

Han E, Nohadani O (2025) Nonlinear decision rules made scalable by nonparametric liftings. *Management Sci.* 71(4):3449–3471.

Kuhn D, Wiesemann W, Georghiou A (2011) Primal and dual linear decision rules in stochastic and robust optimization. *Math. Programming* 130(1):177–209.

Laguna M (1998) Applying robust optimization to capacity expansion of one location in telecommunications with demand uncertainty. *Management Sci.* 44(11-part-2):101–110.

Mohajerin Esfahani P, Kuhn D (2018) Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Math. Programming* 171(1–2):115–166.

Muth JF (1960) Optimal properties of exponentially weighted forecasts. *J. Amer. Statist. Assoc.* 55(290):299–306.

Powell WB (2011) *Approximate Dynamic Programming: Solving the Curses of Dimensionality* (Wiley, New York).

Puterman ML (1994) *Markov Decision Processes: Discrete Stochastic Dynamic Programming* (Wiley, New York).

Rahal S, Papageorgiou DJ, Li Z (2021) Hybrid strategies using linear and piecewise-linear decision rules for multistage adaptive linear optimization. *Eur. J. Oper. Res.* 290(3):1014–1030.

See CT, Sim M (2010) Robust approximation to multiperiod inventory management. *Oper. Res.* 58(3):583–594.

Thomä S, Walther G, Schiffer M (2024) Designing tractable piecewise affine policies for multi-stage adjustable robust optimization. *Math. Programming* 208(1–2):661–716.

Vayanos P, Kuhn D, Rustem B (2011) Decision rules for information discovery in multi-stage stochastic programming. *Proc. 50th IEEE Conf. Decision Control Eur. Control Conf.* (IEEE, Piscataway, NJ), 7368–7373.

Wang Z, Qi M (2020) Robust service network design under demand uncertainty. *Transportation Sci.* 54(3):676–689.

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